Coastal flood exposure of Singapore under extreme sea levels and AR6 sea-level rise

A screening study with open data — technical report

Wenjie Luo (independent personal project; not affiliated with NUS, TMSI or PUB)

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Summary

This report estimates which parts of Singapore would be below, and hydraulically connected to, extreme still-water sea levels today and under IPCC AR6 sea-level rise, and what lies there. It is a screening exercise built entirely on open data; it is **not an official flood map**. Coastal defences, drainage, waves and rainfall are not represented.

- The 100-year still-water level at Tanjong Pagar is 1.90 m above the 1995–2014 mean sea level (95 % CI 1.72–2.15 m), from a GEV fit to 37 detrended monsoon-season maxima.
- Adding AR6 medium-confidence projections gives 100-year levels of 1.94 m (2020) to 2.66 m (SSP5-8.5, 2100).
- Connected flooding grows from 7.9 km² to 13.7 km² (AR6 likely range 11.4–18.3 km²) across those cases; 53% of the 2100 SSP5-8.5 area is on offshore islands. Exposure is small: 1,558 buildings within ~60 m of flooding and a building-weighted proxy of 3,460 residents.
- The dominant uncertainty is vertical: a ±0.3 m change in the geoid-to-MSL offset moves the 2100 SSP5-8.5 area between 10.6 and 18.1 km²; ignoring the offset gives 5.6 km². Hydraulic connectivity also matters: the naive bathtub gives 36.3 km².
- A 2-D local-inertial model of one mainland sub-domain, verified against an analytical solution (RMSE 8.0 cm), floods 98% of the bathtub area under a realistic 72-hour storm-tide hydrograph, and 100% when the peak is held constant.

1. Data

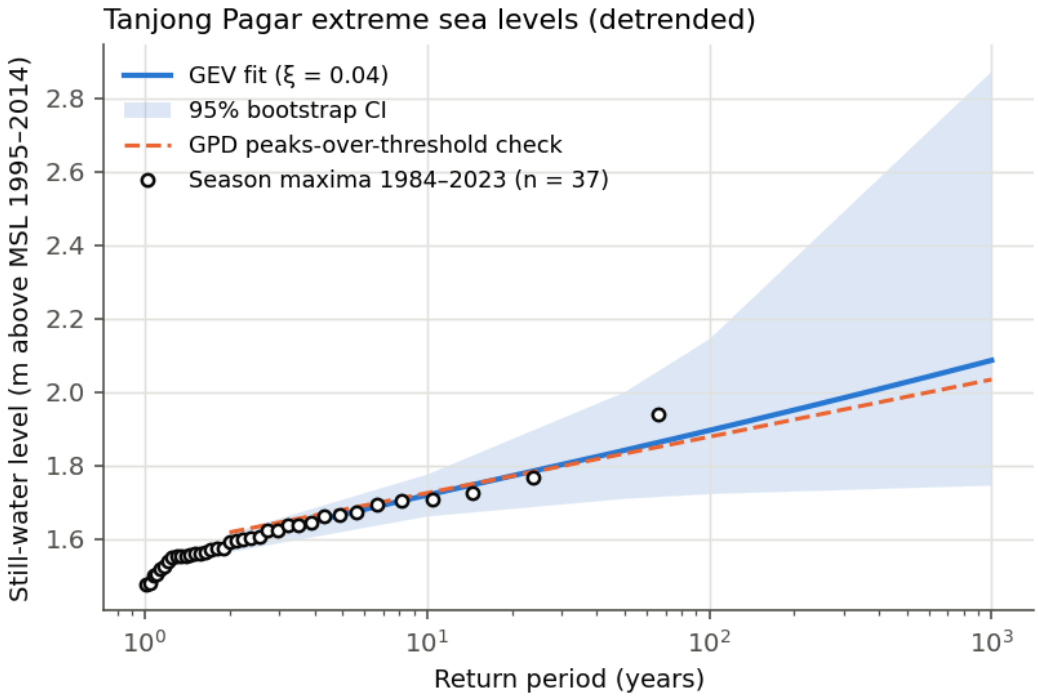
Input	Source	Notes
Hourly sea level	UHSLC station 699, Tanjong Pagar, 1984–2023	338,246 valid hours, chart datum
Sea-level projections	IPCC AR6 (Garner et al. 2021), PSMSL 1746	total RSL incl. VLM, medium confidence
Elevation	Copernicus GLO-30 DSM	1″ (~31 m), EGM2008 heights
Geoid-to-MSL offset	DTU22MDT	1.15 m mean over Singapore Strait cells
Boundaries, population	URA MP2019 planning areas and subzones; SingStat Census 2020	data.gov.sg
Assets	OSM buildings (126,346) and roads (10,108 km); LTA MRT exits (613)	

2. Methods

Reference level. All water levels are metres above the 1995–2014 mean sea level, which is 1.71 m above the Tanjong Pagar chart datum and is also the AR6 baseline, so return levels and projections add directly.

Extremes. The hourly record is detrended by the linear trend of annual means (3.1 ± 0.5 mm/yr) anchored at 2005. Maxima are taken per July–June season (81% fall in November–February); seasons with less than 80 % data are dropped. A GEV is fitted by maximum likelihood (μ = 1.566, σ = 0.065, ξ = 0.04; bootstrap 95 % range of ξ -0.35 to 0.28; Gumbel not rejected, p = 0.70). Return-level intervals come from 1,000 non-parametric bootstrap resamples. Two cross-checks fall inside the interval: a GPD on declustered exceedances of the 99.9th percentile (124 clusters) gives 1.88 m for 100 years, and removing each year's mean instead of a trend gives 1.92 m.

Return period	Level (m)	95 % CI (m)
10 yr	1.72	1.66–1.78
100 yr	1.90	1.72–2.15
1,000 yr	2.09	1.75–2.88



Return levels at Tanjong Pagar

Sea-level rise. AR6 medians and likely ranges (17th–83rd percentiles) at Tanjong Pagar:

Scenario	2050 (m)	2100 (m)
SSP1-2.6	0.18 [0.10–0.27]	0.43 [0.23–0.67]
SSP2-4.5	0.19 [0.12–0.28]	0.54 [0.35–0.81]
SSP5-8.5	0.22 [0.14–0.31]	0.76 [0.55–1.08]

The observed trend implies 4.7 cm of rise from 2005 to 2020, consistent with the AR6 value of 4.6 cm for 2020. The AR6 low-confidence high-end storyline is not included.

Vertical datum. Copernicus heights refer to the EGM2008 geoid, but local mean sea level lies above the geoid by the mean dynamic topography. Heights are converted with z_MSL = z_EGM2008 − 1.15 m (DTU22MDT). With the offset, 20% of the land is below 5 m above MSL, against 13% without it; the official figure is about 30 %, and a surface model should under-count low ground, so the offset is directionally supported but not validated.

Connected inundation. Water cells in GLO-30 are flattened to 0 m. Zero cells linked to the raster edge and outside the land polygons are open sea; zero cells inside the land polygons and exactly flat patches of ≥ 300 cells are regulated inland water (barrage reservoirs), which neither flood nor convey water. A priority flood gives each land cell the minimum still-water level at which it connects to the sea; a scenario floods all cells whose fill level is below its water level.

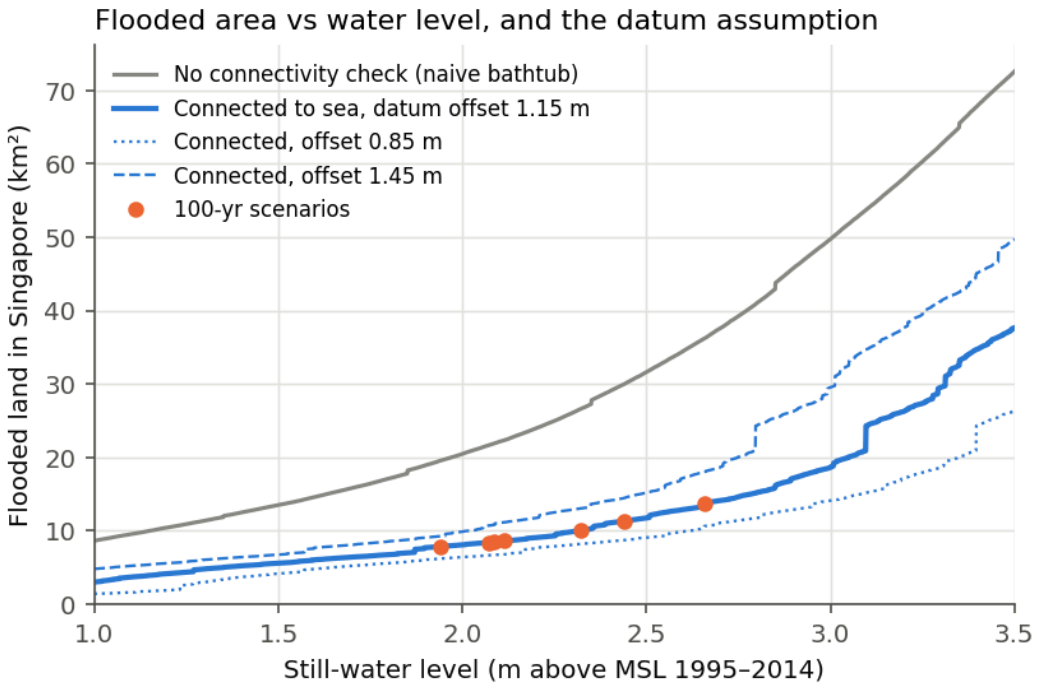
2-D model. The local-inertial equations (Bates et al. 2010) are solved on the 30 m grid of a 180 × 180-cell mainland sub-domain (103.744–103.794°E, 1.255–1.305°N), forced at sea cells by the observed 1999-12-23 storm tide raised so its peak equals 2.66 m. Verification: analytical front test (RMSE 8.0 cm, 3.4% of peak depth), lake-at-rest and mass-conservation tests (largest relative mass error 7e-13).

Exposure. Buildings and MRT exits count if a flooded cell is within ~60 m (the DSM puts roof heights at the building itself); roads count where their own cell floods; residents are Census 2020 subzone populations scaled by the share of exposed buildings.

3. Results

100-yr event	Level (m)	Area (km²) [likely]	Mainland (km²)	Buildings	Residents (proxy)	Roads (km)
2020	1.94	7.9	3.1	941	2,500	69
SSP2-4.5 2050	2.09	8.5 [8.1–9.0]	3.4	1,045	2,720	76
SSP1-2.6 2100	2.32	10.0 [8.7–12.6]	4.4	1,264	3,020	93
SSP2-4.5 2100	2.44	11.3 [9.3–14.1]	5.2	1,344	3,130	104
SSP5-8.5 2100	2.66	13.7 [11.4–18.3]	6.5	1,558	3,460	134

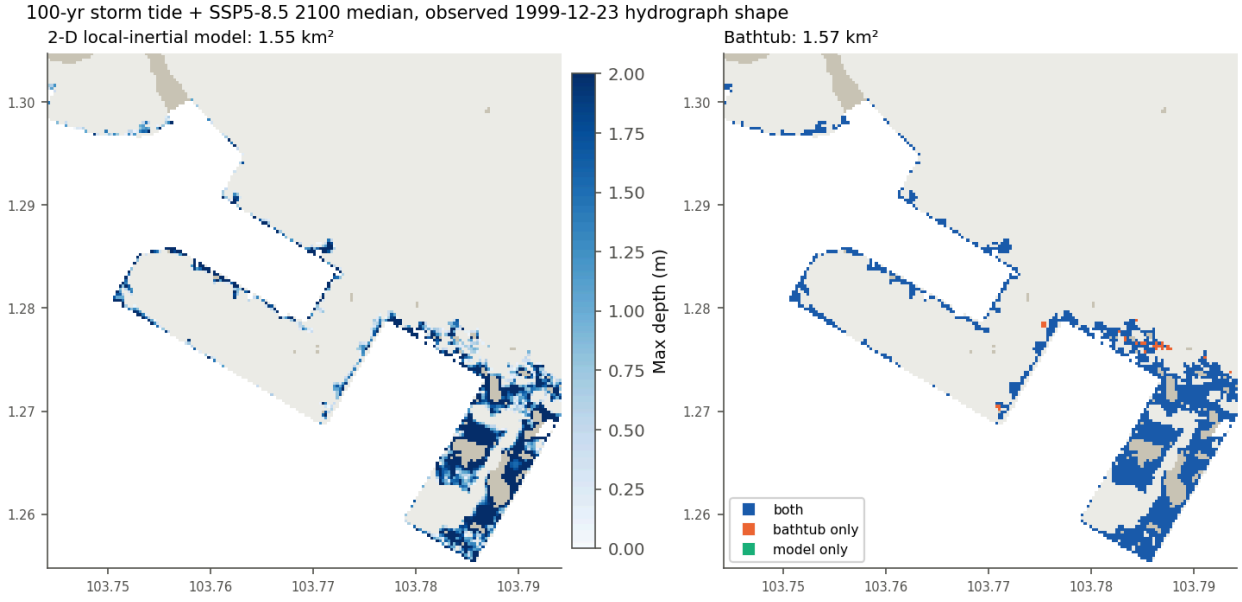
The largest planning-area totals for the 2100 SSP5-8.5 100-yr case are North-Eastern Islands (3.7 km²), Western Islands (3.2 km²), Queenstown (1.6 km²). MRT stations with an exit within ~60 m of flooding in that case: Marina Bay, Marina South, Marina South Pier.



Flooded area against water level and datum offset

Sensitivity (100-yr, SSP5-8.5 2100)	Area (km²)
Central (offset 1.15 m, 4-connected)	13.7
Offset −0.3 m / +0.3 m	10.6 / 18.1
No offset (geoid = MSL)	5.6
8-connected	14.8
No connectivity check	36.3

2-D model. Bathtub 1.57 km²; model 1.55 km² (98%); peak held constant 1.57 km² (100%). Halving the time step changes the area by 0.5%; averaging the DEM to 60 m / 90 m loses 20% / 37% of it (narrow low channels are smoothed away), so the result depends on the 30 m terrain; Manning n from 0.02 to 0.06 gives 1.56 to 1.54 km².



2-D model against the bathtub

4. Limitations

1. GLO-30 is a radar surface model (2011–2015) with metre-level vertical error; buildings and canopy raise apparent ground; radar noise and post-acquisition reclamation distort the coast.
2. The geoid-to-MSL offset comes from a global MDT that is least reliable near coasts.
3. Defences, gates, pumps, drains, waves, rainfall and compound flooding are not modelled.
4. 37 seasons of data constrain a 100-yr level only loosely; stationarity of the surge climate is assumed.
5. Exposure counts are proximity-based proxies, not damage estimates.

References

- Bates, P. D., Horritt, M. S. & Fewtrell, T. J. (2010). A simple inertial formulation of the shallow water equations for efficient two-dimensional flood inundation modelling. *J. Hydrol.* 387, 33–45.
- Barnes, R., Lehman, C. & Mulla, D. (2014). Priority-flood: an optimal depression-filling and watershed-labeling algorithm for digital elevation models. *Comput. Geosci.* 62, 117–127.
- Caldwell, P. C., Merrifield, M. A. & Thompson, P. R. (2015). Sea level measured by tide gauges from global oceans — the Joint Archive for Sea Level holdings. NOAA NCEI, accession 0019568.
- Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer.

- de Almeida, G. A. M., Bates, P., Freer, J. E. & Souvignet, M. (2012). Improving the stability of a simple formulation of the shallow water equations for 2-D flood modeling. *Water Resour. Res.* 48.
- Fox-Kemper, B. et al. (2021). Ocean, cryosphere and sea level change. In *Climate Change 2021: The Physical Science Basis*, IPCC AR6 WGI, ch. 9.
- Garner, G. G. et al. (2021). IPCC AR6 sea level projections. Zenodo, 10.5281/zenodo.6382554.
- Hunter, N. M., Horritt, M. S., Bates, P. D., Wilson, M. D. & Werner, M. G. F. (2005). An adaptive time step solution for raster-based storage cell modelling of floodplain inundation. *Adv. Water Resour.* 28, 975–991.
- Kulp, S. A. & Strauss, B. H. (2019). New elevation data triple estimates of global vulnerability to sea-level rise and coastal flooding. *Nat. Commun.* 10, 4844.
- DTU Space. DTU22MDT mean dynamic topography. ftp.space.dtu.dk/pub/DTU22/MDT.